

Abhishek Teja Goli

Fairborn, Ohio 45324 | (937)-443-7879 | g.abhishekteja@gmail.com | [linkedin.com/in/abhishektejagoli](https://www.linkedin.com/in/abhishektejagoli)

PROFESSIONAL SUMMARY

Data Science graduate specializing in healthcare machine learning, statistical modeling, and generative AI. Built FDA-scale clinical data pipelines and glaucoma risk-scoring models at Topcon Healthcare, with statistical validation work presented at ARVO. Skilled in Python and SQL for ML models, ETL pipelines, and RAG/LLM applications, turning structured and unstructured healthcare data into deployable, validated tools.

WORK EXPERIENCE

Data Science Intern | Topcon Healthcare - Remote

June 2025 – Present

- Developed and validated Logistic Regression-based HT and FN glaucoma risk-scoring tools, engineering TSNIT, Macula, GCL, and RNFL features across 90+ real-world-data sites to automate patient-level risk scoring, replacing manual chart review.
- Engineered an automated PEC data restoration and validation framework processing 45K+ clinical files per location, with dry runs, batch processing, rollback, and audit logging.
- Curated and standardized FDA Workplace Screening datasets spanning FY2016-2025, automating record matching and validation across ~109K healthcare records to improve data quality.
- Built a statistical validation framework using ANOVA, Bland-Altman analysis, and coefficient of variation to evaluate repeatability and reproducibility of three glaucoma risk scores across devices and operators; presented findings at ARVO.
- Built regex-Pandas ETL tools for multi-site log parsing and deduplication, cutting processing time from hours to seconds.
- Worked with Clinical Science team and automated the scripts for multiple business scenarios.

Research Assistant, KASTLE lab | Wright State University – Dayton, Ohio

March 2025 – Present

- Evaluated 5 transformer-based embedding models against 60+ ontology classes using cosine similarity and cross-model consensus, with BERT sentence embeddings achieving the highest mean similarity (0.69), informing model selection for downstream ontology-matching tasks.

Data Analyst Intern | DataBeat – Telangana, India

Feb 2024 – June 2024

- Developed a ResNet-based deep learning model achieving 94% accuracy for leaf disease classification for client Envu, enabling automated crop health screening from field images.
- Built and optimized Python and Azure Data Factory ETL pipelines, reducing data processing time by 50% and improving reliability of downstream reporting.
- Automated ad-report extraction and preprocessing to support forecasting models and improve forecast accuracy.
- Contributed to PoCs extracting text data from PDFs using Tesseract OCR and PyPDF.

PROJECTS

Organization-Specific LLM Application (RaiderGPT) | Tech: Python, LangChain, FAISS, Docker, AWS

- Developed an organization-specific RAG application for secure, context-aware knowledge retrieval.
- Engineered FAISS-based semantic search to retrieve relevant information from 1,000+ URLs.
- Built query-routing and prompt-generation pipelines to improve contextual responses and reduce LLM hallucinations.
- Deployed a Dockerized web application on AWS with load balancing and shared database access for concurrent users.

Smart Movie Recommendation System | Tech: Python, RNN-LSTM, NLP, Collaborative Filtering

- Implemented an NLP-based sentiment classifier for movie tweets across positive, negative, and neutral classes.
- Trained an RNN-LSTM model on movie tweets and deployed it for real-time sentiment inference.
- Implemented hybrid collaborative and content-based filtering, improving recommendation relevance by 35%.

Mental Health Severity Classification System | Tech: Python, Scikit-learn

- Developed a machine learning system for DASS-based mental health severity classification, achieving 92% accuracy.
- Applied PCA, K-fold cross-validation, Decision Tree, Naive Bayes, and KNN models for feature reduction and comparative model evaluation.

EDUCATION

Wright State University - Dayton, OH

Aug 2024 – Present

Master of Science in Data Science | GPA: 3.72/4.0

Mahatma Gandhi Institute of Technology - Telangana, India

Aug 2020 – June 2024

B. Tech in Computer Science and Engineering (Data Science)

TECHNICAL SKILLS

Programming & Data : Python, SQL, Pandas, NumPy, SciPy, ETL, Data Cleaning, Data Engineering

Machine Learning & Statistics : Scikit-learn, TensorFlow, PyTorch, Regression, Classification, Predictive Modeling, Statistical Analysis, Feature Engineering, Model Evaluation

NLP & Generative AI : NLP, LLMs, RAG, Transformers, Embeddings, BERT, FAISS, LangChain, Prompt Engineering

Cloud, Visualization & Deployment: Databricks, AWS, Azure, Databricks, Power BI, Tableau, Flask, Streamlit, Docker, Git

CERTIFICATIONS

- Certified in **Data Science** by IBM.
- Google Data Analytics** from Google
- Earned **Python** and **Machine Learning** certification from **Kaggle**.